

Introduction

Sharks are highly efficient swimmers and possess unique skin features called denticles that may contribute to their hydrodynamic performance. This study investigates how denticle-inspired surface features modify near-wall flow, delay flow separation, and influence loading behavior when structural flexibility is introduced.

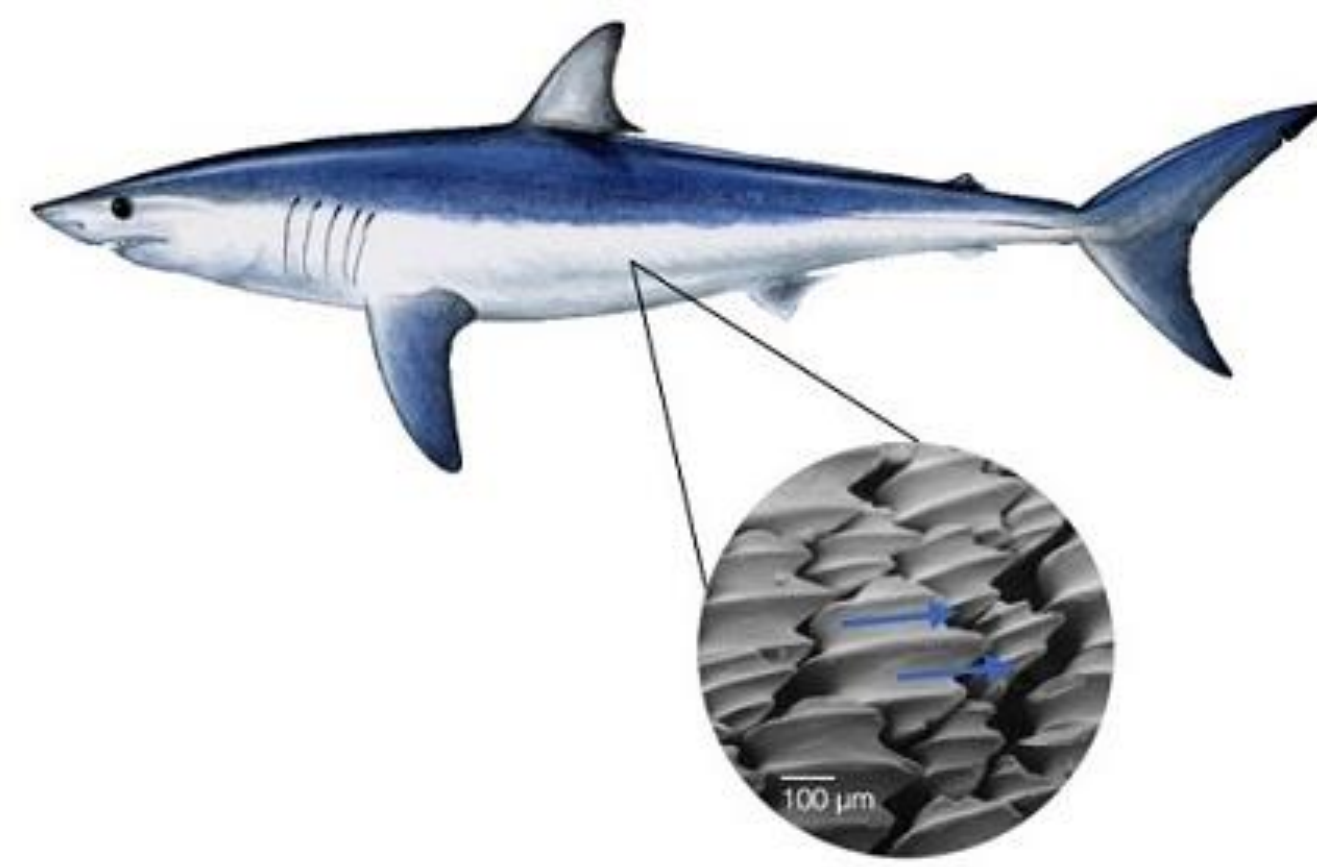


Figure 1. Shark Denticles shown on a shark body

Hypothesis

Denticle-inspired geometry alters the local flow structure near the wall, producing periodic flow separation and recirculating flow regions. When structural flexibility is introduced, these flow modifications are expected to produce nonuniform loading and different structural responses among the elements.

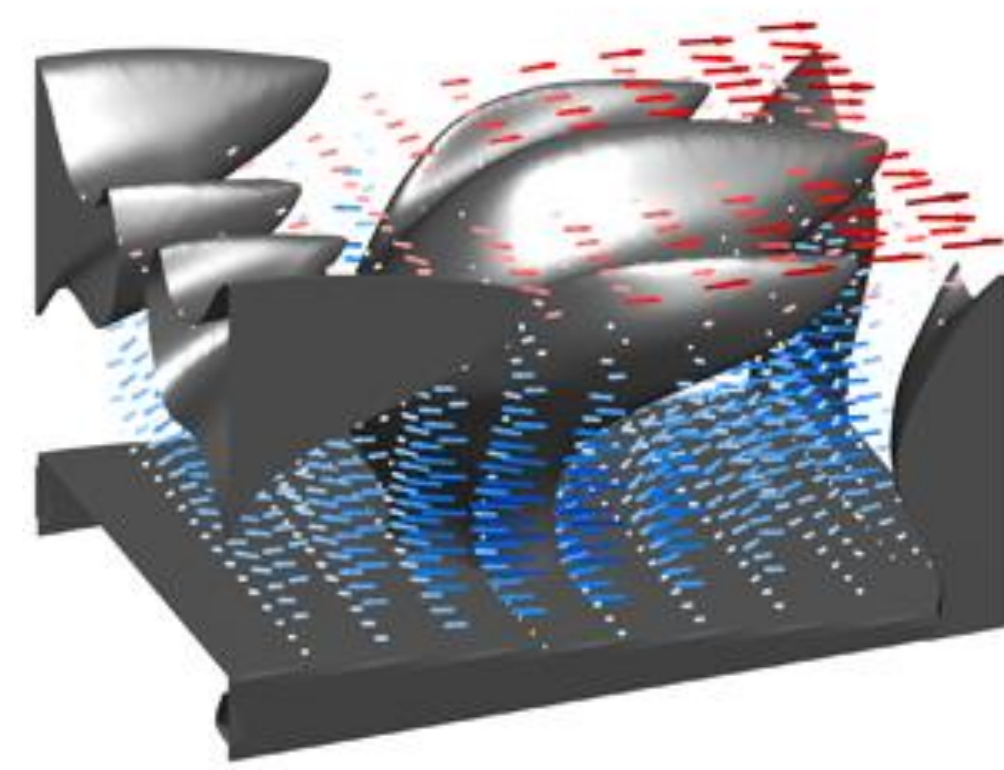


Figure 2. Shark Denticles shown in flow

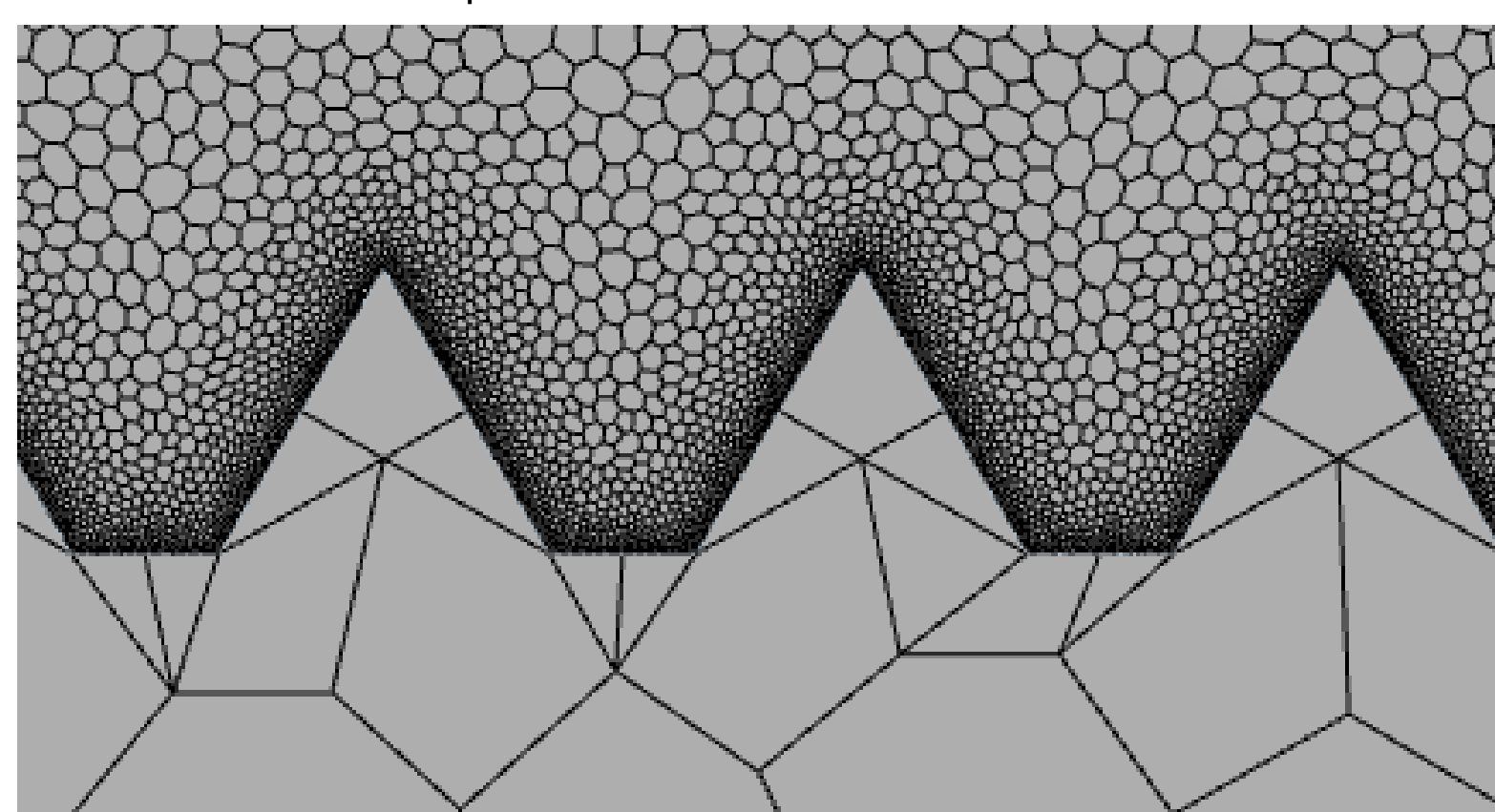
Objectives

- Investigate how simplified denticle-inspired geometry modifies near-wall flow.
- Identify key flow features such as separation, recirculation, and repeating vortical structures.
- Extend the analysis from a rigid 2D model to a 3D FSI model.
- Evaluate how fluid loading varies across multiple elements when structural response is enabled.
- Compare rigid and flexible configurations to better understand the role of coupling between flow and structure.

Methodology

The study first used a 2D rigid CFD model with simplified denticle-inspired surface geometry to isolate the effect of shape on near-wall flow. This model showed that the repeating geometry causes flow separation at the peaks, recirculation within the valleys, and a periodic vortical pattern along the surface, indicating that the denticle features actively modify the boundary layer rather than acting as passive roughness.

Figure 3. Simplified 2D denticle-inspired geometry used to isolate the effect of surface shape on near-wall flow.



While this 2D approach can identify the main flow structures created by the geometry, it does not include bending features as seen in shark denticles.

2D Model Setup

The study first used a 2D rigid CFD model with simplified surface geometry to isolate the effect of shape on near-wall flow. This model showed that the repeating geometry causes flow separation at the peaks, recirculation within the valleys, and a vortical pattern along the surface, showing that the denticle features modify the boundary layer rather than acting as passive roughness. While this 2D approach was somewhat effective, it does not show structural deflection under fluid loading.

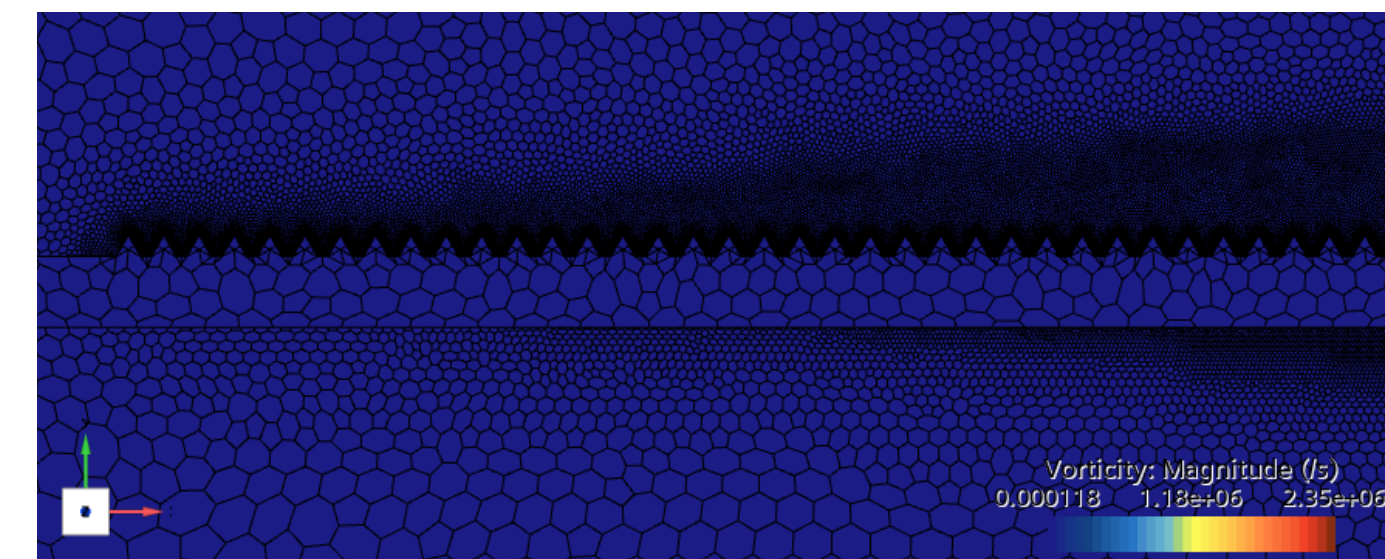


Figure 4. 2D Wake refinement for flow separation analysis

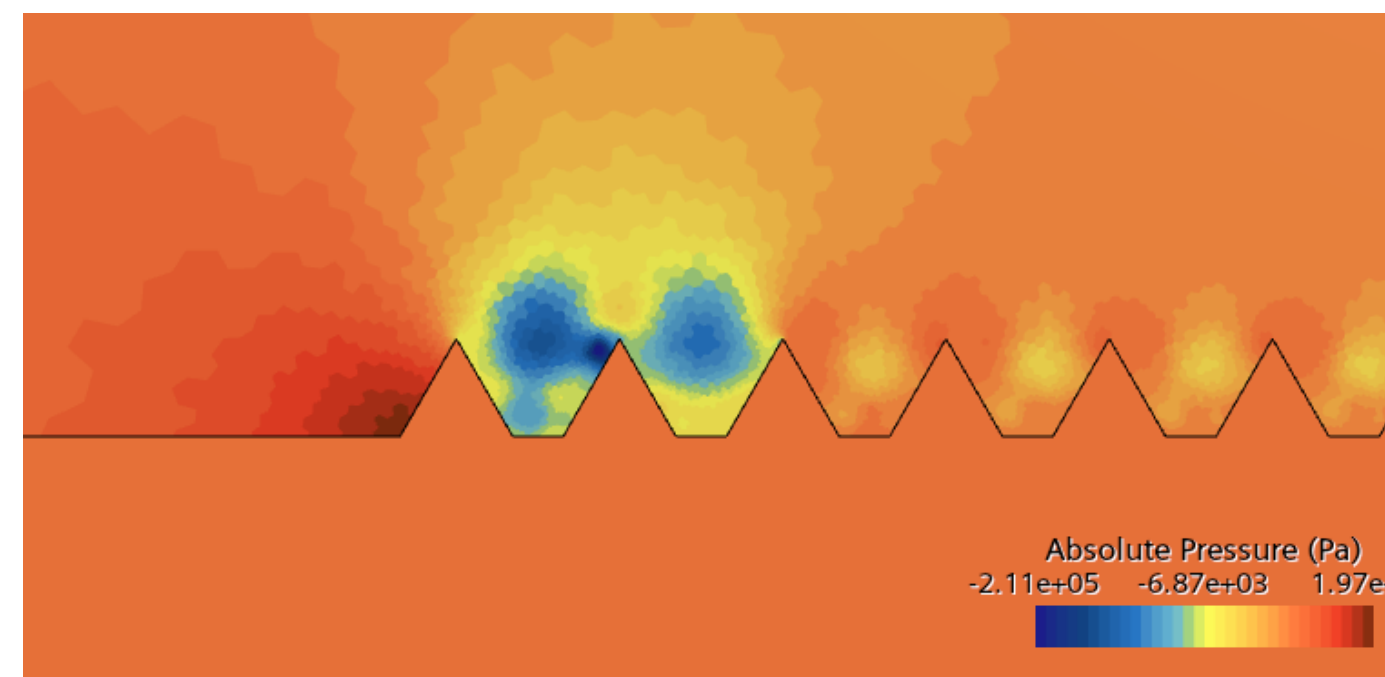


Figure 5. 2D flow-field results showing separation and recirculation generated by the repeating denticle-inspired surface geometry.

FSI Model Setup/Geometry

To capture both flow behavior and structural response, the study was extended to a 3D fluid–structure interaction model:

- 3D geometry (.03m diameter, .043m height, .066m spacing)
- Rectangular domain with inlet velocity 20 m/s (Shark speed)
- Si was used as a material for bending
- Multiple interacting elements
- FSI enabled in STAR-CCM+
- Structural response allowed under fluid loading
- Mesh refined near the surface features

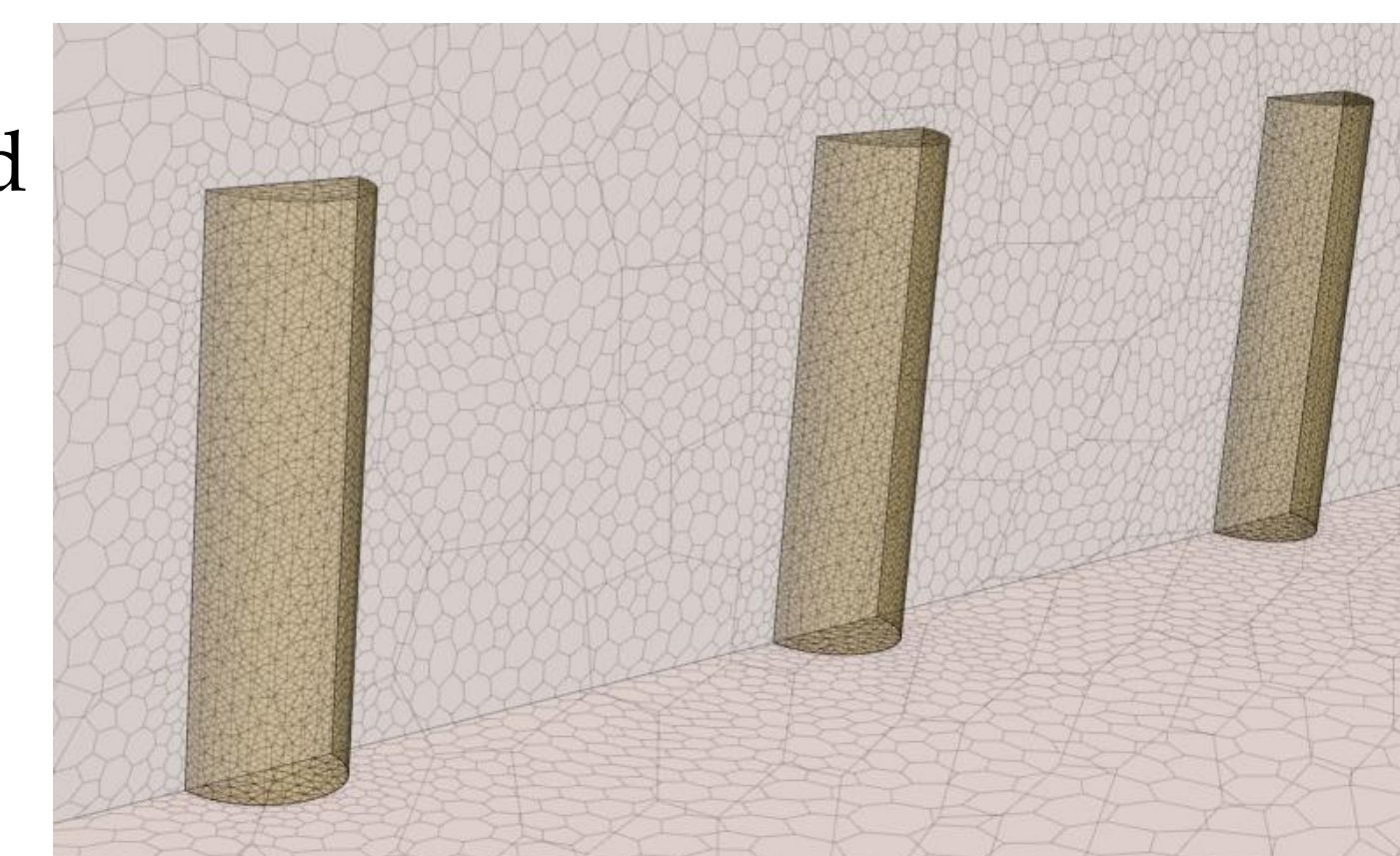


Figure 6. 3D denticle-inspired model used for fluid–structure interaction analysis.

This model allows the structures to respond to the surrounding flow, making it possible to evaluate how local flow features influence force distribution and deformation.

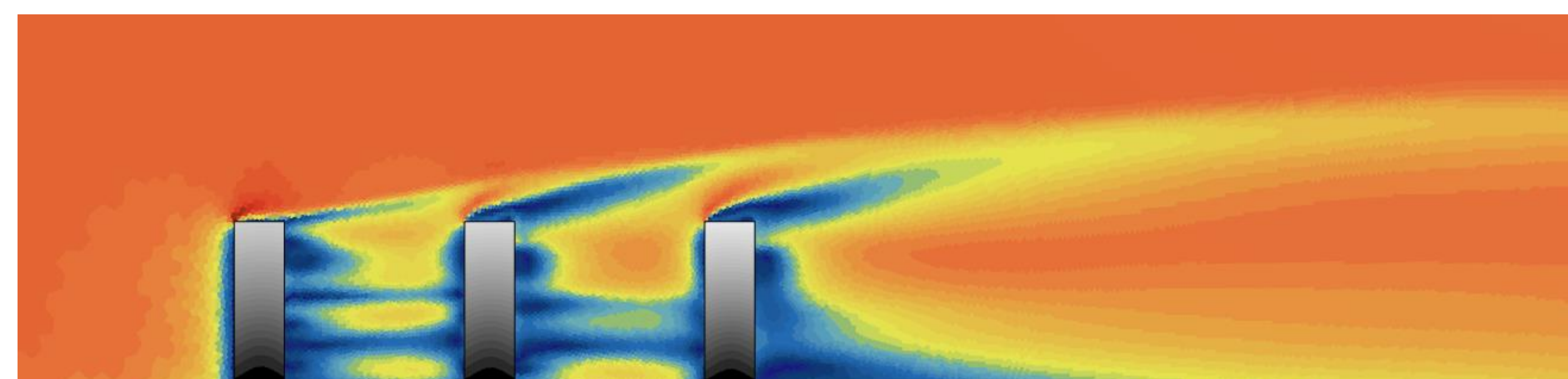


Figure 7. Initial flow characteristics of FSI Denticles

The 3D FSI model shows that the flow and loading are not uniform across the surface features.

Observed behavior

- Local pressure varies from one element to another
- Vortical structures develop around the features
- Different elements experience different loading conditions
- Structural displacement is not uniform across the array
- These results suggest that when flexibility is introduced, the interaction between geometry and flow becomes more complex. Rather than experiencing identical conditions, individual elements respond differently depending on their local flow environment.

Velocity and Vorticity Comparison

Metric	FSI (Mean ± Std)	Rigid (Mean ± Std)	Percent Difference (%)
Velocity Magnitude (m/s)	15.50 ± 4.39	15.46 ± 4.39	-0.26
Velocity i (m/s)	14.99 ± 5.18	14.95 ± 5.18	-0.24
Velocity j (m/s)	0.41 ± 2.78	0.40 ± 2.77	-2.16
Vorticity k (1/s)	-369.30 ± 2536.58	-368.22 ± 2509.04	-0.29

Figure 8. Table above compares between Velocity and Vorticity for a stationary and FSI – bending case of Shark Denticles

Key observations:

- Mean velocity differences between cases are minimal
- Vorticity magnitudes are comparable between cases
- This suggests the flow difference between cases is minimal

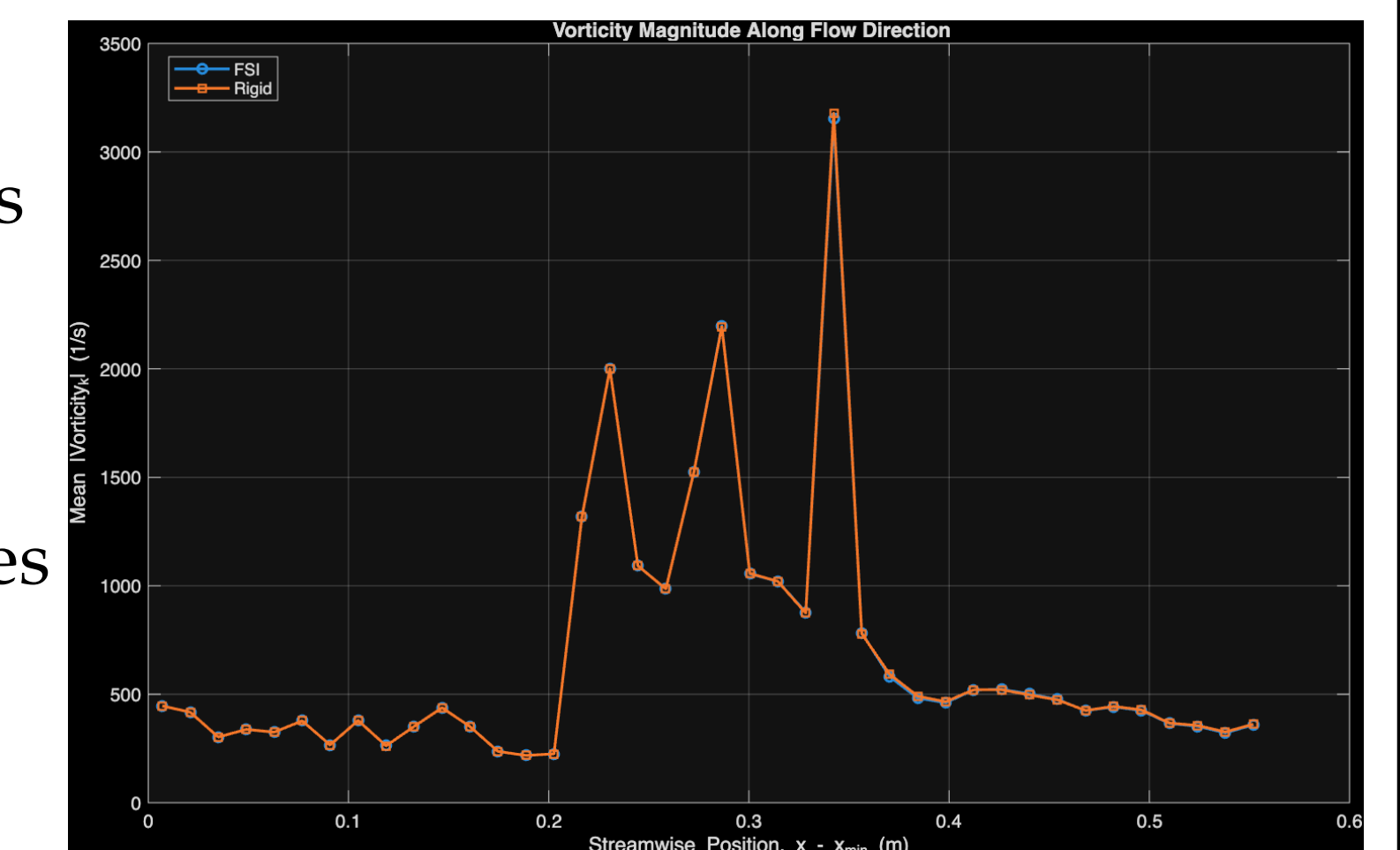


Figure 9. Comparison of velocity magnitude and vorticity magnitude for the rigid and FSI denticle cases, where the largest changes happen near the denticle region.

Denticle Force and Displacement

Denticles experience loading driven by flow structures such as separation and recirculation. The FSI results show that these structures lead to displacement and force responses, particularly in the first parts in the incoming flow. The interaction between upstream and downstream denticles show that wake effects influence structural loading.

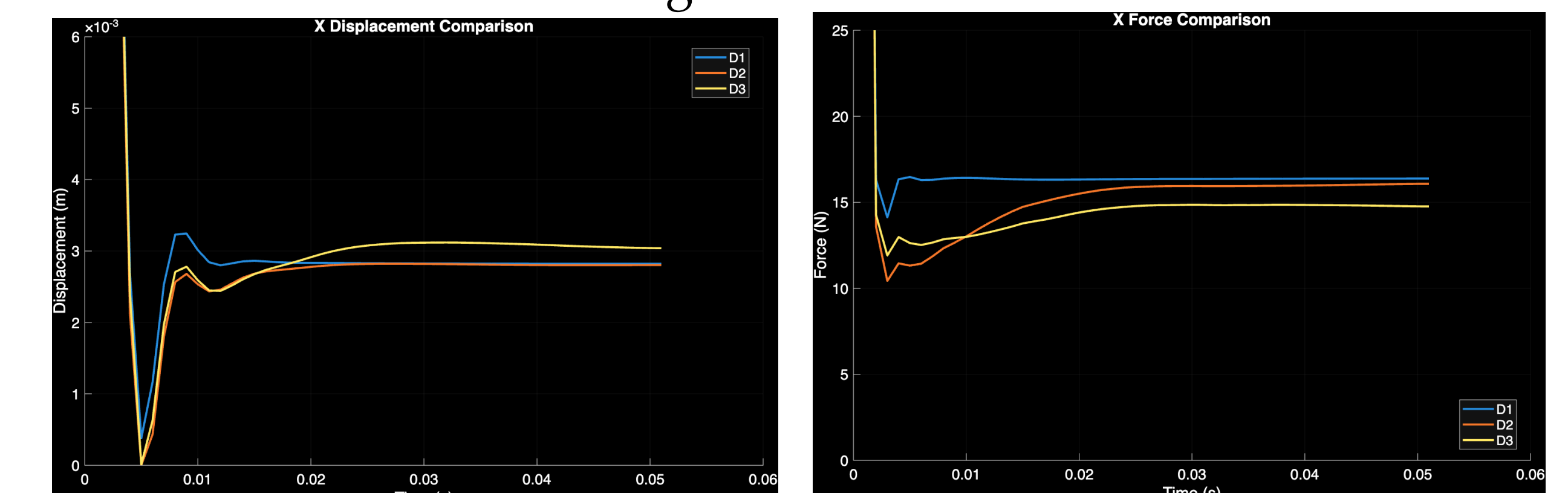


Figure 10. X Displacement and Force values shown on the denticles with flow initiated.

Conclusions

- Both the 2D and 3D results show that surfaces with denticle-like features influence both the fluid field and structural loading behavior.
- The 2D rigid model demonstrates that geometry alone generates organized flow features such as separation and recirculation.
- The 3D FSI model extends this by showing that these flow features lead to non-uniform and time-dependent structural response.

These findings indicate that denticle-inspired designs may influence performance not only with the flow control, but also through FSI coupling effects that change different loading conditions.

- “From Shark Skin to Speed,” Illumin, University of Southern California.
- Journal of Fluid Mechanics, “Thrust Generation by Shark Denticles.”
- STAR-CCM+ documentation, Siemens Digital Industries Software.